Spam Email/SMS Classifier

Project Overview

This project is a **Spam Email/SMS Classifier** developed using **Python** and **Machine Learning**. It classifies a given SMS or email message as **Spam** or **Not Spam (Ham)**.

The project uses the **SMS Spam Collection Dataset** from Kaggle and the **Multinomial Naive Bayes** algorithm with **TF-IDF Vectorizer** for text classification.

Requirements

Before running the project, install the following software:

- Python 3.10 or later
- Visual Studio Code (Recommended)

Python Libraries

Install the required libraries using:

```
py -m pip install pandas numpy scikit-learn nltk joblib
```

How to Run the Project

Step 1

Download or copy the complete project folder to your computer.

Step 2

Open the project folder in **Visual Studio Code**.

Step 3

Open the terminal in VS Code.

Step 4

Install the required libraries (only the first time):

```
py -m pip install pandas numpy scikit-learn nltk joblib
```

Step 5

Run the project using:

```
py spam_classifier.py
```

How to Use

1. Run the program.
2. Wait for the model to load and display the accuracy.
3. When prompted, enter any SMS or email message.
4. The program will display one of the following:
5. **SPAM**
6. **NOT SPAM**
7. Type **exit** to close the program.

Example

Input

```
Congratulations! You have won ₹50,000. Click here to claim your prize.
```

Output

```
Prediction: SPAM
```

Input

```
Hello, today's class starts at 10 AM.
```

Output

```
Prediction: NOT SPAM
```

Dataset

- **Dataset Name:** SMS Spam Collection Dataset
- **Source:** Kaggle

https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset

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Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- NLTK
- Joblib
- TF-IDF Vectorizer
- Multinomial Naive Bayes

Notes

- Keep **spam.csv** in the same folder as **spam_classifier.py**.
- Do not rename **spam_model.pkl** or **vectorizer.pkl**.
- If Python is installed but `pip` does not work, use:

```
py -m pip install pandas numpy scikit-learn nltk joblib
```

- If you modify the dataset, run the program again to train and save a new model.

Author

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